ClaimSathi — Software Requirements Specification (Build Day MVP)

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1. Overview

ClaimSathi takes 3 PDFs (policy, rejection letter, discharge summary + bill) and in under 90 seconds returns a verdict on whether a rejected Indian health insurance claim breaks the policy or IRDAI rules, with cited clauses and a ready-to-send appeal letter.

Problem. In FY25, 8% of 3.26 crore health claims were rejected (about 1 in 12). In a LocalCircles survey, 36% of policyholders said their claim was fully rejected for invalid reasons. Most people never appeal because they cannot read a 40-page policy against a rejection letter.

Pitch line. Opus investigates, Sonnet fights for you.

Build Day goal. A working web app that runs the 4 demo cases end to end, live, in front of judges.

Success criteria (must all be true at demo time)

1. Case 01 returns STRONG CASE with the moratorium violation, citing Clause 6.4.
2. Case 04 returns VALID REJECTION. This proves the tool is honest.
3. Investigation streams step by step on screen. Nothing sits on a blank spinner for more than 5 seconds.
4. One click produces an appeal letter, and one more click produces it in Hindi.
5. A cached fallback plays the full demo even if Wi-Fi or the API fails.

2. Scope

Build only what the 2-minute demo shows. Everything else is a slide line, not code.

Priority	Feature	Status at demo
P0	Upload 3 PDFs (policy wording + schedule, rejection letter, discharge/bill)	Must work
P0	Opus investigation with streamed checklist	Must work
P0	Verdict screen: badge, violations, clause quotes, amount at stake	Must work

Priority	Feature	Status at demo
P0	Sample-case buttons (load Case 01–04 in one click)	Must work
P0	Cached demo mode (replay saved results, no API)	Must work
P1	Appeal letter to insurer's Grievance Officer (Sonnet)	Should work
P1	Hindi translation of the letter	Should work
P1	Escalation timeline with computed dates	Should work
P2	Ombudsman complaint draft	Nice to have
P2	"Ask your policy" chat	Nice to have
P2	PDF viewer with clause highlight	Nice to have (fallback: quoted clause card)

Out of scope: login, database, payments, scanned/handwritten OCR, insurer integrations, mobile app, real patient data.

Cut order if behind schedule: chat → Ombudsman draft → PDF highlight → Hindi → timeline. Never cut P0.

3. Users and user flow

The primary user is a policyholder or family member holding a rejection letter, with no insurance or legal knowledge.

User	Need	What ClaimSathi gives
Policyholder (primary)	"Was my rejection fair? What do I do?"	Verdict, reasons in plain words, letter to send
Family member (often the one who files)	Act fast, in Hindi or Marathi	Translated letter, deadlines
Hackathon judge (demo)	See real reasoning, not a chatbot	Live streamed checklist with clause citations

Happy path (one straight run):

1. User lands on the home screen and taps **Try sample case 01** (or uploads 3 PDFs).

2. App sends PDFs to `/api/investigate`. The investigation screen streams checks as they complete.
3. Verdict screen shows the badge, 1–5 violations each with a policy quote and an IRDAI rule, and the amount at stake.
4. User taps **Draft appeal letter**. Sonnet writes it in about 10 seconds.
5. User taps **हिंदी में** to translate, then **Copy** or **Download .txt**.
6. User sees the escalation timeline: insurer (30 days) → Bima Bharosa → Ombudsman (within 1 year, free, up to ₹50 lakh).

4. Functional requirements

Each requirement is done only when its acceptance test passes on the demo dataset.

ID	Requirement	Acceptance test	Priority
FR-1	Upload up to 4 PDFs (policy wording, schedule, rejection letter, discharge + bill), each ≤ 10 MB, drag-drop or picker	All 4 Case 01 files accepted; a .docx shows "PDF only"	P0
FR-2	Sample-case buttons load Case 01–04 files from `backend/data/samples`	One click starts Case 01 with no upload	P0
FR-3	Send PDFs to Opus 5.5 as base64 `document` blocks with `irdai_rules.json` in the system prompt	Request succeeds; no OCR code	P0
FR-4	Stream progress events (`step`, `fact`, `check`, `verdict`) to the UI via Server-Sent Events	First event on screen within 5 s	P0
FR-5	Extract key facts: first inception date, current period, sum insured, PEDs declared, admission/discharge date-time, diagnosis, procedure, amount claimed, rejection reason, clause cited	Case 01 facts match the answer key	P0
FR-6	Compute dates in code, not by the model: months of continuous cover, stay length in hours	Case 01 = 85 months; Case 02 = 7 h 40 m	P0

ID	Requirement	Acceptance test	Priority
FR-7	Run every applicable rule in `irdai_rules.json` and return pass / fail / not-applicable with evidence	Case 03 flags 48 vs 36 months	P0
FR-8	Return verdict: STRONG CASE / CONTESTABLE / VALID REJECTION + confidence (high/medium/low)	All 4 cases match `expected_verdicts.json`	P0
FR-9	Every violation shows an exact quote from the policy PDF with page number	Quote text found in the PDF	P0
FR-10	Show amount at stake, capped by sum insured and sub-limits	Case 02 = ₹48,500 (under ₹60,000 cataract cap)	P0
FR-11	Demo mode: replay saved JSON results with realistic delays when `?demo=1` or API fails	Works with Wi-Fi off	P0
FR-12	Draft grievance letter (Sonnet 5.5) addressed to the Grievance Redressal Officer, citing claim no., clauses and rules	Letter names Clause 6.4 for Case 01	P1
FR-13	Translate letter to Hindi (and Marathi if time)	Hindi output renders in Devanagari	P1
FR-14	Escalation timeline with dates computed from the rejection date	Case 01: insurer reply due 20-Sep-2026	P1
FR-15	Ombudsman complaint draft	Includes policy no., claim no., relief sought	P2
FR-16	Chat with your policy (Sonnet, PDFs cached)	Answers "Is cataract covered?" with clause	P2
FR-17	Disclaimer on every result: "Helps you understand and draft. Not legal advice."	Visible on verdict screen	P0

5. IRDAI rules engine

Nine rules live in `data/irdai_rules.json` and go into the Opus system prompt; date math for R1–R3 also runs in Python so the numbers on screen are always right.

ID	Rule	Fails when	Source
R1	Moratorium: after 60 months of continuous cover, no claim can be contested for non-disclosure or misrepresentation except established fraud	Rejection reason = non-disclosure AND months ≥ 60 AND no fraud alleged	IRDAI Insurance Products Regs 2024; policy Clause 6.4
R2	Pre-existing disease waiting capped at 36 months	Letter applies > 36 months, or months ≥ 36 for a declared PED	IRDAI Products Regs 2024; Excl01
R3	Specified disease waiting capped at 36 months (policy says 24)	Letter applies a longer period, or the period is complete	Excl02
R4	24-hour rule does not apply to Day Care procedures	Rejection for stay < 24 h AND procedure in Day Care list	Clause 1.5, 2.2, Annexure I
R5	Every repudiation needs Claims Review Committee approval	Letter does not mention it	IRDAI Health Master Circular 2024; Clause 5.4
R6	Rejection must cite the specific policy clause	Letter cites a clause name only, no number	Master Circular 2024; Clause 5.4
R7	Insurer must tell the user about grievance / Ombudsman options	Not mentioned	Clause 5.4
R8	Insurer collects documents from the hospital	Rejected for "documents not submitted"	Master Circular 2024; Clause 5.2
R9	Sub-limits apply; payable = min(bill, sub-limit, sum insured)	Used for amount at stake only	Policy Section 4

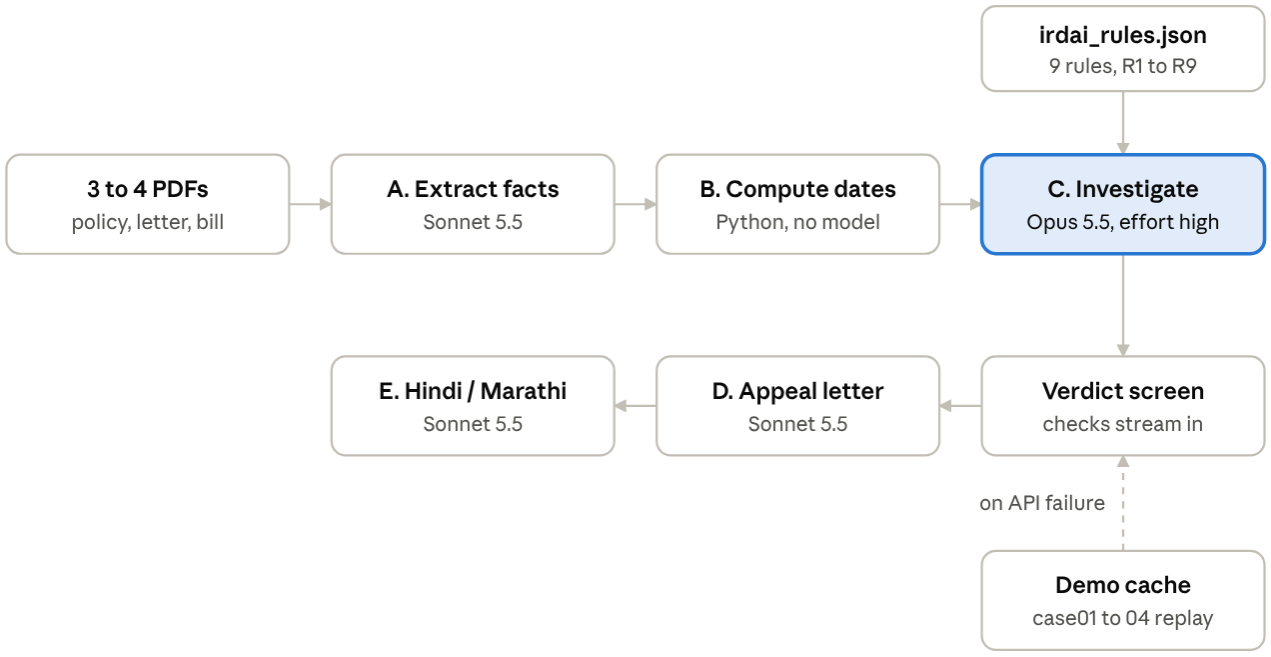
Escalation constants: insurer reply 30 days → Bima Bharosa → Ombudsman within 1 year of final reply, free, claims up to ₹50 lakh, insurer must comply within 30 days or pay ₹5,000/day.

```
{
  "id": "R1",
  "name": "Moratorium period",
  "applies_when": "rejection_category == 'non_disclosure'",
  "check": "months_continuous_cover >= 60 && !fraud_alleged",
  "on_fail": "Claim cannot be contested for non-disclosure after 60 months of
continuous cover.",
  "policy_clause": "6.4",
  "regulation": "IRDAI (Insurance Products) Regulations, 2024",
  "severity": "critical"
}
```

6. AI architecture

Three model calls run in sequence on the server. Each one emits an SSE event, so the UI moves at every stage. Date math runs in code between calls 1 and 2.

Opus investigates in the middle; Sonnet reads and writes around it



ClaimSathi pipeline · 5 stages + demo fallback

Rules feed only Opus; the dashed demo cache takes over whenever a live call fails.

Stage	Model	Input	Output	Target time
A. Extract facts	`claude-sonnet-5-5`	3–4 PDFs as `document` blocks	`Facts` (Pydantic, FR-5)	⩽ 20 s
B. Compute	Python (`services/dates.py`)	facts	months of cover, stay hours, deadlines	< 50 ms
C. Investigate	`claude-opus-5-5` , effort `high`	PDFs (cached) + facts + computed values + `irdai_rules.json`	`Investigation` (checks + verdict)	⩽ 60 s
D. Draft letter	`claude-sonnet-5-5`	investigation + facts	letter text (English)	⩽ 15 s
E. Translate	`claude-sonnet-5-5`	letter	Hindi / Marathi letter	⩽ 15 s

API rules that prevent failures

- PDFs go as `{"type":"document","source":{"type":"base64","media_type":"application/pdf","data":"..."}}` with `cache_control` on the last document, so stages C–E reuse the cache.
- Get JSON with structured outputs or strict tool use (`strict: true` , `tool_choice: auto`). Sonnet 5.5 returns HTTP 400 for forced `tool_choice` `any` or `tool` .
- Validate every JSON response with Pydantic. On a parse failure, retry once; on a second failure, switch to demo mode for that case.
- Set a 90-second timeout per call. On a timeout, show the cached result labelled "replay".
- The UI reveals each check with a 400 ms stagger after stage C returns, so results are real and the pacing is readable.

Stage C system prompt (core)

```
You are ClaimSathi's Claim Investigator for Indian health insurance.
You receive: the policy wording, policy schedule, rejection letter and
discharge summary/bill,
plus extracted facts, computed values (trust these numbers) and IRDAI rules
(JSON).
For EVERY rule: decide pass | fail | not_applicable, quote the exact policy
text with page number,
and explain in one plain sentence a non-lawyer can follow.
Never invent clauses, dates or amounts. If evidence is missing, say "not
found".
If the rejection is correct, say so clearly: verdict VALID_REJECTION.
Return only JSON matching the schema.
```

Investigation schema (Pydantic)

```
from typing import Literal, Optional
from pydantic import BaseModel

class Check(BaseModel):
    rule_id: str                        # "R1"
    title: str                          # "Moratorium period"
    status: Literal["pass", "fail", "not_applicable"]
    finding: str                        # one plain sentence
    policy_quote: Optional[str] = None
    policy_clause: Optional[str] = None # "6.4"
    page: Optional[int] = None
    regulation: Optional[str] = None
    severity: Literal["critical", "major", "minor"]

class Investigation(BaseModel):
    verdict: Literal["STRONG_CASE", "CONTESTABLE", "VALID_REJECTION"]
    confidence: Literal["high", "medium", "low"]
    headline: str                       # "Rejection breaks the 5-year
moratorium rule"
    amount_at_stake_inr: int
    checks: list[Check]
    next_steps: list[str]

# usage: Investigation.model_validate_json(text)
# JSON schema for the model: Investigation.model_json_schema()
```

Date math (stage B)

```
from dateutil.relativedelta import relativedelta

def months_of_cover(first_inception: date, admission: date) -> int:
    d = relativedelta(admission, first_inception)
    return d.years * 12 + d.months      # Case 01: 2019-06-15 -> 2026-08-03 =
85

def stay_hours(admit: datetime, discharge: datetime) -> float:
    return (discharge - admit).total_seconds() / 3600   # Case 02: 7.67
```

7. Tech stack, structure and API

Backend is Python only: FastAPI serves every API and model call; Next.js is a UI-only client. No database: uploads live in memory by case ID, and demo results are static JSON files.

Layer	Choice	Why
Backend	Python 3.11+, FastAPI, Uvicorn	Async, auto API docs at `/docs`
AI	`anthropic` Python SDK (`AsyncAnthropic`)	Opus 5.5 + Sonnet 5.5
Validation	Pydantic v2	Safe JSON from models
Streaming	`sse-starlette` (`EventSourceResponse`)	Server-Sent Events in a few lines
Uploads	`python-multipart` (`UploadFile`)	Multipart PDF upload
Date math	`python-dateutil` (`relativedelta`)	Exact months of cover
Config	`python-dotenv`	`ANTHROPIC_API_KEY` in `backend/.env`
Frontend	Next.js 15 + TypeScript + Tailwind + shadcn/ui + framer-motion	UI only, no API routes
CORS	FastAPI `CORSMiddleware`, allow `http://localhost:3000`	Frontend on 3000, backend on 8000
Run	`uvicorn main:app --reload --port 8000` + `npm run dev`	Local laptop; no deploy risk

Folder structure

```
claimsathi/
  backend/
    main.py                    # FastAPI app, CORS, include routers
    routers/
      cases.py                 # POST /api/cases
      investigate.py           # GET  /api/investigate/{case_id}  (SSE)
      letters.py               # POST /api/letter, /api/translate
    services/
      claude.py                # AsyncAnthropic client, model ids, PDF blocks
      prompts.py               # system prompts A, C, D, E
      dates.py                 # months of cover, stay hours, deadlines
      demo.py                  # replay cached results
      store.py                 # in-memory {case_id: files, facts,
investigation}
    models/schemas.py          # Pydantic models
    data/
      irdai_rules.json
      demo/case01.json ... case04.json
      samples/case01/*.pdf ... case04/*.pdf
    requirements.txt
    .env                       # ANTHROPIC_API_KEY=...
  frontend/
    app/page.tsx                          # Screen 1
    app/investigate/[caseId]/page.tsx     # Screen 2 (EventSource)
    app/result/[caseId]/page.tsx          # Screen 3
    components/ UploadZone CheckItem VerdictBadge ViolationCard LetterPanel
Timeline
    lib/api.ts                            #
NEXT_PUBLIC_API_URL=http://localhost:8000
```

requirements.txt

```
fastapi
uvicorn[standard]
anthropic
pydantic>=2
sse-starlette
python-multipart
python-dateutil
python-dotenv
```

API routes

Two-step flow because the browser's `EventSource` only supports GET: upload first, then open the stream.

Route	Method	Body	Returns
`/api/cases`	POST (multipart)	`files: list[UploadFile]` or form field `sample_case=case01`	`{ "case_id": "..." }`
`/api/investigate/{case_id}`	GET (SSE)	query `demo=0|1`	events: `step`, `facts`, `computed`, `check`, `verdict`, `error`
`/api/letter`	POST (JSON)	`{ case_id, type: "grievance" | "ombudsman" }`	`{ letter }`
`/api/translate`	POST (JSON)	`{ text, lang: "hi" | "mr" }`	`{ text }`
`/api/health`	GET	none	`{ ok, models }` (test before demo)

SSE endpoint skeleton

```
@router.get("/api/investigate/{case_id}")
async def investigate(case_id: str, demo: int = 0):
    async def events():
        if demo:
            async for e in demo_replay(case_id):
                yield e
            return
        try:
            yield {"event": "step", "data": "reading"}
            facts = await extract_facts(case_id)              # stage A,
Sonnet
            yield {"event": "facts", "data": facts.model_dump_json()}
            computed = compute(facts)                          # stage B,
Python
            yield {"event": "computed", "data": computed.model_dump_json()}
            yield {"event": "step", "data": "checking"}
            inv = await investigate_claim(case_id, facts, computed)  # stage
C, Opus
            for c in inv.checks:
```

```
                yield {"event": "check", "data": c.model_dump_json()}
                await asyncio.sleep(0.4)
            yield {"event": "verdict", "data": inv.model_dump_json()}
        except Exception:
            async for e in demo_replay(case_id, label="replay"):
                yield e
    return EventSourceResponse(events())
```

SSE event example

```
{"type":"check","data":{"rule_id":"R1","status":"fail","title":"Moratorium
period","finding":"Policy held 85 months; non-disclosure cannot be used after
60."}}
```

8. UI and UX

Three screens, one accent colour, and one memorable moment: the checklist turning red line by line.

Design tokens: background `#FAFAF7`, ink `#1A1A1A`, accent indigo `#3B3BD6`, fail red `#D93025`, pass green `#188038`, warning amber `#E8A200`. Font: Inter (UI), Noto Sans Devanagari (Hindi). Cards: 12 px radius, 1 px border, no heavy shadows. Laptop-first at 1440 px; must not break at 375 px.

Screen 1 — Home / Upload

- Headline: "Your claim was rejected. Let's check if that was fair." Sub-line: "1 in 12 health claims in India is rejected. Find out in 60 seconds."
- Three labelled drop zones: Policy (wording + schedule), Rejection letter, Hospital papers. Each turns green with the file name on upload.
- Primary button: **Investigate my claim** (disabled until 3 zones are filled).
- Row of 4 sample-case chips: "Heart attack ₹2.37L", "Cataract ₹48.5K", "BP emergency ₹1.31L", "Knee replacement ₹3.02L".

Screen 2 — Investigation (the wow moment)

- Left (60%): vertical checklist. Each row = icon (spinner → ✓ green / ✗ red / – grey), rule title, one-line finding. Rows slide in with framer-motion.
- Right (40%): "Facts we found" card filling live: patient, policy since, months of cover (big number), stay length, amount claimed, rejection reason.
- Top: stage stepper — Reading documents → Calculating → Checking IRDAI rules → Verdict.

Screen 3 — Verdict and action

- Hero badge: STRONG CASE (green), CONTESTABLE (amber), VALID REJECTION (grey) + headline + "₹2,37,000 at stake".
- Violation cards: rule title, plain-language finding, quoted clause in a grey quote block with "Policy p. 3, Clause 6.4", regulation tag.
- Action bar: **Draft appeal letter**, **हिंदी में**, **Ombudsman complaint**, **Copy**, **Download**.
- Timeline strip: Today → Insurer reply due (+30 days) → Bima Bharosa → Ombudsman deadline (+1 year).
- For VALID REJECTION: calm tone, "This rejection appears correct" + what to do next (keep renewing; claim payable after waiting period ends on a computed date).
- Footer on every result: "Helps you understand and draft. Not legal advice."

9. Non-functional requirements

The demo must survive bad Wi-Fi, a slow API and a nervous presenter.

ID	Requirement	Target
NFR-1	Time to first visible progress	⩽ 5 s
NFR-2	End-to-end investigation (Case 01)	⩽ 90 s live; ⩽ 25 s in demo mode
NFR-3	Verdict accuracy on the 4 demo cases	4 / 4 match `expected_verdicts.json`
NFR-4	Resilience	Any API error or timeout falls back to cached result with a small "replay" label
NFR-5	Cost per investigation	Under $0.50 (prompt caching on PDFs)
NFR-6	Privacy	No files written to disk or logged; API key only in `backend/.env`, never in client code
NFR-7	Honesty	Never invents clauses; shows "not found" when evidence is missing; disclaimer on every result
NFR-8	Accessibility	Status shown by icon + text, not colour alone; contrast ⩾ 4.5:1
NFR-9	Language	English UI; letters in English, Hindi (P1), Marathi (P2)

10. Build plan and fail-safes

Arrive with the skeleton running and demo JSON saved; spend build time only on the AI pipeline and polish.

Before the event (tonight)

- [ ] Create Organization at console.anthropic.com, generate API key, test one call to `claude-opus-5-5` and `claude-sonnet-5-5`
- [ ] Backend: `python -m venv .venv && source .venv/bin/activate && pip install -r requirements.txt`, then `uvicorn main:app --reload --port 8000` and open `/docs`. Frontend: `npx create-next-app@latest frontend --ts --tailwind --app`, `npx shadcn@latest init`, `npm i framer-motion lucide-react`
- [ ] Copy the 4 demo cases into `backend/data/samples/case01..04/`
- [ ] Write `backend/data/irdai_rules.json` (Section 5) and `backend/models/schemas.py` (Section 6)
- [ ] Build the 3 static screens with hard-coded Case 01 data (no AI yet)
- [ ] Run stages A + C once per case from a script; save outputs to `data/demo/case0X.json` (this is the fail-safe)
- [ ] Commit to GitHub; charge laptop; download fonts offline; set up phone hotspot

At the event

Block	Build	Done when
0:00–0:30	Wire upload + sample chips to `/api/investigate`; FastAPI SSE plumbing with fake events	Fake checklist animates
0:30–1:30	Stage A (Sonnet extract) + `dates.py` + stage C (Opus investigate)	Case 01 returns STRONG CASE live
1:30–2:00	Run all 4 cases; tune the prompt until 4/4 match	`expected_verdicts.json` matches
2:00–2:30	Verdict screen polish + letter (stage D) + Hindi (stage E)	Letter + Hindi in ⩽ 30 s
2:30–2:45	Timeline strip; demo-mode toggle; error fallback	Demo works with Wi-Fi off
2:45–3:00	Rehearse the 2-minute script twice; freeze code	No more commits

Fail-safes

Risk	Fallback
Wi-Fi drops or API slow	`?demo=1` replays saved JSON with the same animation
Model returns bad JSON	Pydantic catches it → one retry → cached result
Verdict wrong on stage	Use demo mode for that case; keep the live run for Case 01 only
Running out of time	Follow the cut order in Section 2; P0 alone is a complete demo
Laptop dies	Code on GitHub; recorded 60-second screen capture on phone

11. Test plan and demo script

The build passes when all 4 demo cases match the answer key; Case 01 and Case 04 are the ones shown live.

Case	Claim	Must find	Expected verdict
01 Rajesh Mehta	Heart attack, ₹2,37,000	R1 fail (85 months > 60), R5 fail (no committee), R6 fail (vague clause), R7 fail (no Ombudsman)	STRONG CASE
02 Sunita Patil	Cataract, ₹48,500	R4 fail (Day Care DC-01); R3 pass (44 months > 24); amount ₹48,500	STRONG CASE
03 Vikram Joshi	BP emergency, ₹1,31,000	R2 fail (letter says 48, policy says 36; held 46 months); R7 fail	STRONG CASE
04 Anita Desai	Knee replacement, ₹3,02,000	R3 pass for insurer (6.5 months < 24); R5, R6, R7 pass	VALID REJECTION

Extra checks: upload a .docx (rejected cleanly); turn Wi-Fi off (demo mode plays); upload a real Arogya Sanjeevani policy wording (no crash, sensible facts).

2-minute demo script

1. **(0:00–0:15) Hook.** "1 in 12 health claims in India is rejected. 36% of people say theirs was rejected unfairly. Most never appeal."

2. **(0:15–1:05) Live run, Case 01.** Click "Heart attack ₹2.37L". Narrate as rows turn red: "85 months of cover, and the moratorium rule says that after 60 months they can't reject for non-disclosure." Verdict: STRONG CASE, ₹2,37,000 at stake.
3. **(1:05–1:30) Action.** Draft appeal letter → हिंदी में → show the Ombudsman deadline.
4. **(1:30–1:45) Honesty.** Load Case 04: "When the insurer is right, ClaimSathi says so." VALID REJECTION.
5. **(1:45–2:00) Close.** "Opus investigates, Sonnet fights for you. Insurance Samadhan proved people pay for claim help — we make it instant for every Indian."

Sources

- Insurance Business: FY25 health claim data
- Business Today: LocalCircles survey and Insurance Samadhan
- BasuNivesh: IRDAI 2024 health insurance changes
- Freefincal: the 24-hour rule
- Ditto: Insurance Ombudsman process
- What's new in Claude Sonnet 5.5